

Evaluating the Impact of Macroeconomic Variables on Machine Learning Models for Loan Default Prediction

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Abstract—This study evaluates whether macroeconomic indicators improve machine learning models used for loan default prediction. LendingClub borrower data were combined with macroeconomic data from the Federal Reserve Economic Data database. Logistic Regression, Random Forest, and XGBoost models were compared using accuracy, precision, recall, F1-score, and ROC-AUC. XGBoost achieved the highest accuracy, precision, and ROC-AUC, while Logistic Regression produced similar overall results. Additionally, an ablation analysis found that adding the macroeconomic variables produced little in terms of predictive value, while borrower-level characteristics were the main drivers of model performance. These findings suggest that model complexity and additional macroeconomic information do not necessarily produce meaningful improvements in loan-default prediction.

Index Terms—Loan default prediction, machine learning, XGBoost, Logistic Regression, Random Forest, macroeconomic indicators, LendingClub, FRED

I. INTRODUCTION

With the increasing use of artificial intelligence and machine learning in banks and financial institutions, lenders believe that these tools can be used to predict loan defaults. When financial institutions experience an increase in credit demand, they often rely on machine learning methods to predict whether a person will default on their loans before extending a loan. Although there have been numerous studies that analyze borrower information such as income, DTI, loan amount, and credit history, few studies actually ask whether the economy affects whether someone repays a loan. This led to the topic of this research, which is analyzing how various macroeconomic factors affect loan default prediction as well as which machine learning model performs best for loan default prediction. Macroeconomic variables such as unemployment rate, Federal Funds Rate, inflation, and home price index were collected from the Federal Reserve Economic Data (FRED) database. These variables were combined with data from LendingClub, which includes variables such as loan and investor data, borrower financials, credit history, and payment and status tracking. Additionally, three different machine learning models were trained in Python to evaluate the results. These machine learning models were Logistic Regression, Random Forest, and XGBoost. The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC in order to see which machine learning algorithm performs best and would be most useful for predicting loan default. This leads us to the main research question of the study. This research attempts to answer the question of whether incorporating macroeconomic

indicators from the FRED database helps improve loan default prediction rather than simply using borrower-level information alone. If these macroeconomic variables are shown to help predict loan default more accurately, financial institutions may be able to improve credit-risk assessment while minimizing losses from borrowers who default. In order to directly test this, an ablation study was conducted by systematically removing variables, including the macroeconomic ones, and measuring the change in performance to analyze which variables are the most important.

II. LITERATURE REVIEW

A. Evolution of Credit Scoring

Prior to the creation of newer machine learning models such as XGBoost and Random Forest, Logistic Regression was the predominant model with regard to credit scoring. The paper Neural Network Credit Scoring Model by David West examines the topic of how neural networks can be used for credit scoring models as opposed to the traditional model of Logistic Regression. West [1] states that Logistic Regression has been the dominant predictor when it comes to loan default due to its simplicity and interpretability. In his study, he finds that neural networks generally produce slightly higher predictive accuracy. However, even though his findings show that there is only a slight improvement when using neural networks, West argues that even small improvements in accuracy can help reduce the amount of money that lenders lose due to loan default. The author concludes that these credit forecasting algorithms should progress and transition into more modern machine learning models. Additionally, another paper that looks into the evolution of credit scoring is the paper Explainable AI Models for Credit Risk Evaluation in Banking by Anumandla Mukesh. Mukesh's [2] paper examines how explainable AI has been adopted by banks for credit-lending systems and how there is a need for these predictive models to be as accurate as possible. The paper examines the challenges of black-box AI models, in which the inputs and outputs are visible but the internal decision-making process is not. This creates challenges in terms of fairness, and Mukesh argues that this highlights the need for increased AI governance and regulation. The main takeaway from this study is that with the increasing adoption of AI and machine learning in credit-lending systems, there needs to be a balance between transparency, accuracy, and precision. These papers are relevant to this study because they allow us to understand

how AI is evolving and the need to balance both fairness and accuracy. Additionally, these papers highlight the need for more variables to be included in these models because they can provide a more complete picture of each borrower and loan. This is why it is important to examine how macroeconomic variables may affect loan-default prediction.

B. XGBoost for Credit Scoring

One machine learning model that has been shown to perform strongly in the literature is the XGBoost model. This model builds decision trees sequentially and attempts to correct errors made by previous trees. One paper that examines this model in relation to credit scoring is A novel XGBoost extension for credit scoring class-imbalanced data combining a generalized extreme value link and a modified focal loss function. This research, done by Jonah Mushava and Michael Murray [3], states that credit-based datasets are highly imbalanced in that there are far more fully-paid loans in the dataset than there are defaulted loans. This creates an issue in the XGBoost model as the model will assume that most people will pay their loan, which may lead to defaulted loans not being detected. Their study proposes a modified XGBoost system that can improve ROC-AUC and help detect more defaulted loans. Another paper that examines the XGBoost model is Evaluating the Impact of Machine Learning and Alternative Data on Credit Precision and Fair Lending in Emerging Markets. This paper by Mohammud Tofeeq [4] compares the models of Logistic Regression, Random Forest, and XGBoost. In his study, he finds that XGBoost provides the strongest predictive performance, and also finds that alternative data can improve model accuracy. These points highlight the trade-off that exists between prediction and fairness. Finally, the paper Study on a Prediction of P2P Network Loan Default Based on the Machine Learning LightGBM and XGBoost Algorithms According to Different High Dimensional Data Cleaning compares the models of XGBoost and LightGBM. Their study [5] found that LightGBM outperforms XGBoost by about 1.28 percentage points. This paper is highly relevant to this study as it also uses LendingClub data. In the authors' discussion they estimate that the improvement could help reduce loan defaults by approximately \$117 million. These papers help highlight the XGBoost model as a lending credit-scoring model. The prior research in these three studies emphasizes how the model can be improved algorithmically. This study examines how macroeconomic variables can impact these models and if they can lead to more accurate predictive outcomes.

C. Machine Learning in Alternative Data

Another important topic in the use of machine learning in credit algorithms is the specific use of machine learning in alternative data. One paper that highlights this is the paper Role of Artificial Intelligence and Alternative Data in Modern Credit Scoring Models: Transforming Lending in India by Shubh Srivastava. Srivastava [6] argues that traditional credit data has limitations and that alternative data is useful because it expands available borrower information. His study [6] also

notes that alternative data can help increase financial inclusion. However, even with these advantages, some concerns arise regarding fairness, as alternative data may contain sensitive private information, which could be a security issue. Furthermore, another piece of research that examines this is the thesis When All Data is Credit Data: Consumer Credit Markets, Technological Development, and Distributive Justice by the author Nikita Aggarwal. In this thesis [7], Aggarwal states that while alternative data can be used to help improve credit prediction and can expand access for borrowers with a limited credit history, it can also introduce bias and discrimination because it may contain private information. Aggarwal [7] concludes by stating that while alternative data may be a productive tool for credit prediction, it requires responsible use and oversight to prevent it from being harmful. Finally, the paper Debiasing Alternative Data for Credit Underwriting Using Causal Inference by Chris Lam further explores this topic of alternative data in machine learning. Lam [8] explains that alternative data may contain a proxy bias in which people of minority status may be discriminated against due to algorithmic unfairness. His study explores how causal inference can be a way to reduce discrimination while maintaining high predictive performance. All of these papers show how, while alternative data can improve credit prediction, there are some concerns in terms of fairness that arise when using this data. This study attempts to reduce this concern by using macroeconomic data relating to the broader economy rather than individual-level alternative data. So while the macroeconomic data being used does fit under the umbrella of alternative data, the hope is that it can help provide insightful information about loan defaults without directly relying on individual-level sensitive information.

D. Explainable AI in Credit Scoring

With the growing use of AI in credit scoring and banking systems, it is important for there to be transparency and for people to understand how exactly these models work. One paper that examines this issue is the paper The Double Edged Sword of AI Big Data: A Sociotechnical Examination of Fairness in Alternative Lending by Savia Dine Kim. This paper [9] examines how AI has been used to improve lending efficiency and prediction. However, the paper [9] states that AI also has the potential to introduce bias and discrimination with its algorithms, which is a problem that needs to be addressed in the future. Kim [9] argues that there needs to be a balance between the growth in AI and machine learning techniques while also continuously evaluating the ethical implications of these systems. The next paper that examines these issues is the paper Investigating the Risks of Algorithmic Bias and Explainability Failures in Credit Risk Models by Imran Hussein Shah. Shah [10] looks at how black-box models reduce decision transparency and how this is an issue because lenders and borrowers will not understand exactly how or why lending decisions are made by financial institutions. Additionally, Shah [10] describes that biases in these systems can undermine fair lending, and one way to counteract this is by using explainability to improve algorithmic accountability. Furthermore,

Shah [10] states that because of these developments in AI and machine learning, especially with regard to black-box models, governance and regulation have become increasingly essential, and that increased transparency could lead to greater stakeholder trust.

E. Research Gap

All of these papers examine the development and future use of AI and machine learning with regard to credit lending systems. These papers show that as these systems become increasingly used in financial institutions, it is important that accuracy and transparency are balanced. This study examines how these models are affected by the inclusion of macroeconomic data and whether this inclusion improves or reduces predictive performance.

III. METHODOLOGY

A. Research Design

This study evaluates whether the FRED macroeconomic variables affect loan default prediction. This is done by analyzing both baseline and enhanced feature sets to evaluate which variables affect the machine learning algorithms. Additionally, this study evaluates three different machine learning models: Logistic Regression, Random Forest, and XGBoost. After training all three of these models in Python, the predictive performance of all three models is analyzed and compared to determine which are best suited for loan-default prediction.

B. Data Collection and Preprocessing

For this study, two datasets were merged. The first dataset is the LendingClub dataset, which contains historical peer-to-peer (P2P) loan data. For this study, loans that were ongoing or had unresolved payment statuses were removed. The final LendingClub dataset contained around 90,000 loans. In order to define the target variable, the loans were classified using a binary 0/1 target. Fully paid loans were classified as 0, and charged-off loans were classified as 1. Additionally, a macroeconomic dataset from the Federal Reserve Economic Data (FRED) was created and merged with the LendingClub dataset. This dataset included variables found on FRED such as Unemployment Rate, Federal Funds Rate, CPI, GDP, Personal Disposable Income, Personal Savings Rate, House Price Index, and Consumer Credit. In order to integrate this data with the LendingClub data, we had to match both datasets by issue date, merge borrower-level and macroeconomic variables, and create baseline and enhanced datasets. In terms of preprocessing, missing numeric values were imputed using the median, while missing categorical values were imputed using the most frequent value. Additionally, relevant loan statuses were filtered, categorical variables were encoded, and the data were split into training, validation, and test sets using a 60:20:20 split.

C. Machine Learning Models

The three machine learning models examined in this study are Logistic Regression, Random Forest, and XGBoost. Logistic Regression, as mentioned in previous literature, is the model that is commonly used for loan-default prediction. This model serves as a baseline linear classification model that has high interpretability and is widely used as a benchmark for credit scoring. The Logistic Regression model was implemented using the scikit-learn library with L2 regularization, the default lbfgs optimizer, a regularization parameter of $C = 1.0$, and a maximum of 1000 training iterations. On the other hand, Random Forest employs an ensemble learning approach with decision trees. This allows it to capture non-linear relationships and reduce overfitting through the use of multiple decision trees. The Random Forest model was implemented using the scikit-learn library with 100 decision trees, a random state of 42, and $n_jobs=-1$ to utilize all available processor cores. All remaining hyperparameters were left at their default values. Finally, the XGBoost model uses a gradient boosting algorithm that captures complex feature interactions. This model serves as a strong benchmark for structured tabular data. The XGBoost model was implemented using the XGBoost library with 1000 estimators, a learning rate of 0.1, a maximum tree depth of 4, and a random state of 42. Early stopping of 30 rounds was used with the validation set, and the best-performing iteration was used for the final evaluation.

D. Model Evaluation

To evaluate the credit-lending performance of each model, several evaluation metrics are used. These metrics include accuracy, precision, recall, F1 score, and ROC-AUC. Accuracy is determined by the percentage of correctly identified loans and helps provide a measure of overall performance for the model. Precision measures how many predicted defaults were actually defaults, which is important because low precision would result in more borrowers being incorrectly classified as likely to default. Recall measures how many actual defaults were correctly identified. The F1 score balances both precision and recall, and the ROC-AUC measures the model's ability to distinguish between fully paid and charged-off loans. Additionally, confusion matrices are used to provide a visual summary of model performance beyond the individual evaluation metrics.

IV. RESULTS

A. Exploratory Data Analysis

Figure 1 illustrates the overall distribution of loan amounts. The figure shows that most loans fall between approximately \$5,000 and \$20,000. Loans below this range are more common than loans above it, suggesting that the distribution is concentrated primarily among small- to moderate-sized loans.

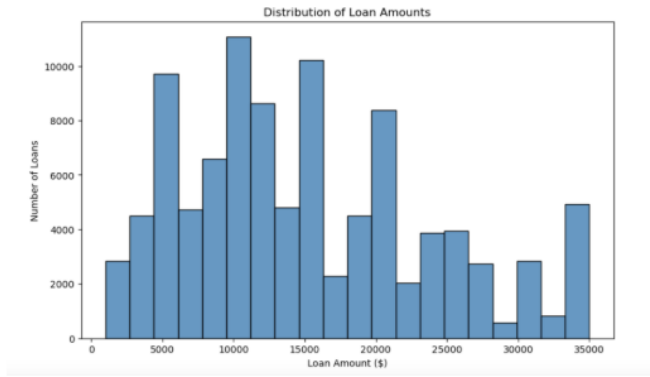


Fig. 1. Distribution of Loan Amounts

Figure 2 examines the interest rates that are assigned to borrowers in the study. On this distribution, the figure shows that most of the interest rates are concentrated between approximately 7% and 18%. Given this information, we can also notice that the graph is highly skewed, as high-interest loans are relatively rare.

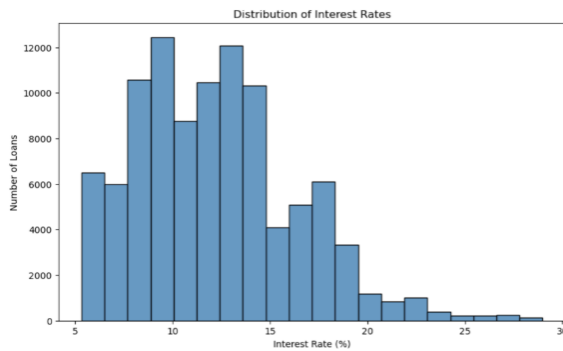


Fig. 2. Distribution of Interest Rates

Figure 3 is a bar graph that examines the loan outcome distribution before unresolved loan statuses were removed from the dataset. The graph shows that a majority of loans were fully paid, while other categories such as current, charged-off, late, and in-grace-period loans were less common. This distribution highlights the class imbalance present in the original LendingClub data and helps explain why the final modeling sample was restricted to fully paid and charged-off loans.

Figure 4 is a correlation heatmap that analyzes the relationship between the borrower and macroeconomic variables. In the heatmap, borrower variables are weakly correlated with macroeconomic indicators. Additionally, the Federal Funds Rate, CPI, and House Price Index are closely related, highlighting that several of the macroeconomic variables capture related broader economic conditions.

Figure 5 captures the interest rate by loan grade. This graph shows that as loan grades worsen from A to G, interest rates rise. Additionally, there is a strong relationship between LendingClub loan grade and borrower risk, which suggests that loan grade is likely an important predictor.

Fig. 3. Loan Status Distribution

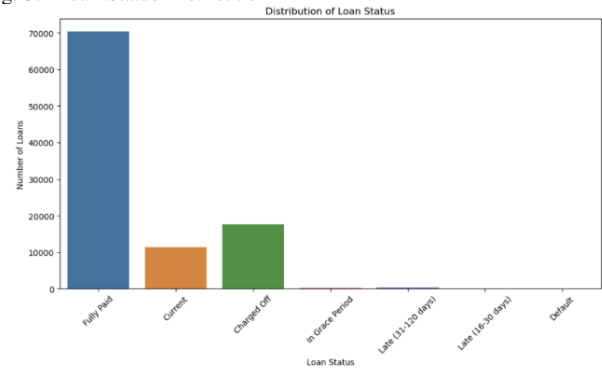


Fig. 4. Correlation Heatmap

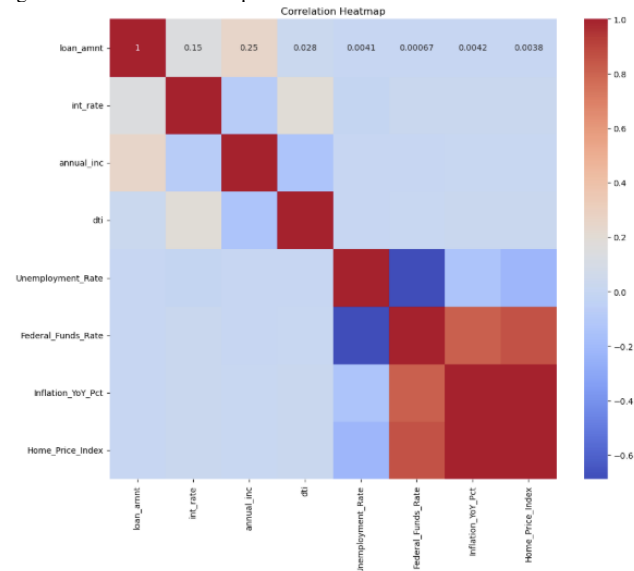
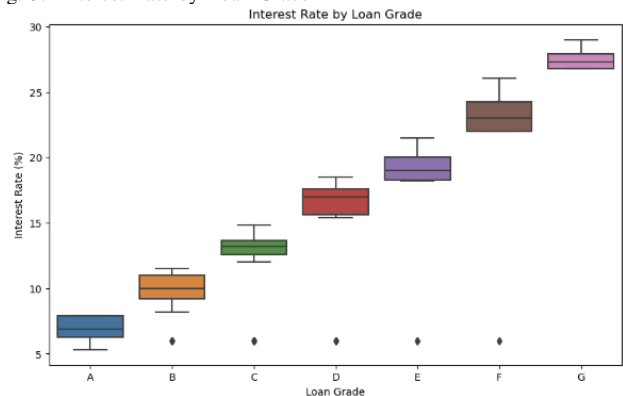


Fig. 5. Interest Rate by Loan Grade



B. Model Performance Comparison

TABLE I
MODEL PERFORMANCE COMPARISON

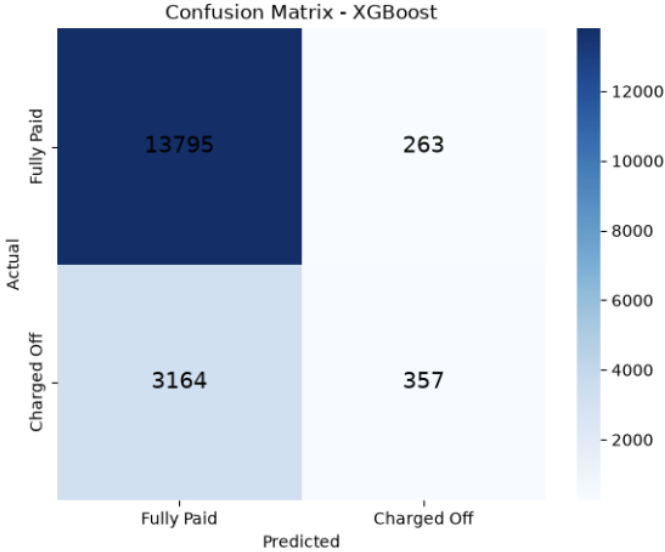
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.803	0.556	0.083	0.145	0.721
Random Forest	0.795	0.460	0.143	0.218	0.684
XGBoost	0.805	0.576	0.101	0.172	0.724

This table compares the predictive performance across all three machine learning models using the evaluation metrics discussed in the methodology. XGBoost achieved the highest accuracy, precision, and ROC-AUC, while Logistic Regression produced similar overall results despite its simpler model structure. Random Forest had the lowest ROC-AUC but achieved the highest recall and F1 score. All three models had relatively low recall scores, suggesting that identifying charged-off loans remained challenging, which is consistent with the class imbalance in the dataset.

C. XGBoost Performance Evaluation

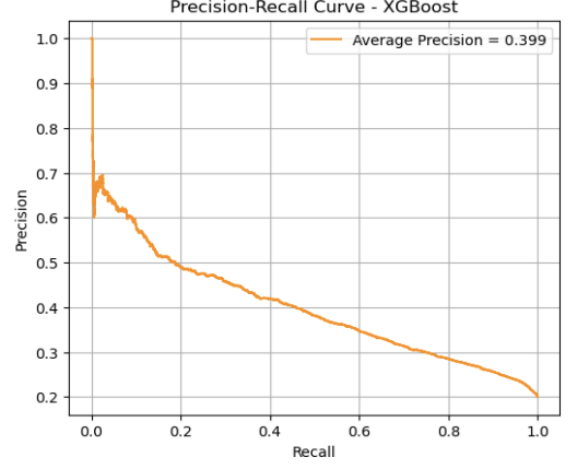
Figure 6 presents the confusion matrix for the XGBoost model.

Fig. 6. XGBoost Confusion Matrix



The confusion matrix shows that the model correctly classified most fully paid loans but struggled to identify charged-off loans. Although XGBoost achieved the highest accuracy, precision, and ROC-AUC, the relatively low recall indicates that many defaults were still classified as fully paid. This is likely due to the class imbalance present within the LendingClub dataset.

Fig. 7. Precision-Recall Curve



The Precision-Recall Curve evaluates minority-class performance. The curve shows an average precision of approximately 0.4. It also illustrates the trade-off between precision and recall, as higher recall is generally associated with lower precision.

Fig. 8. ROC Curve

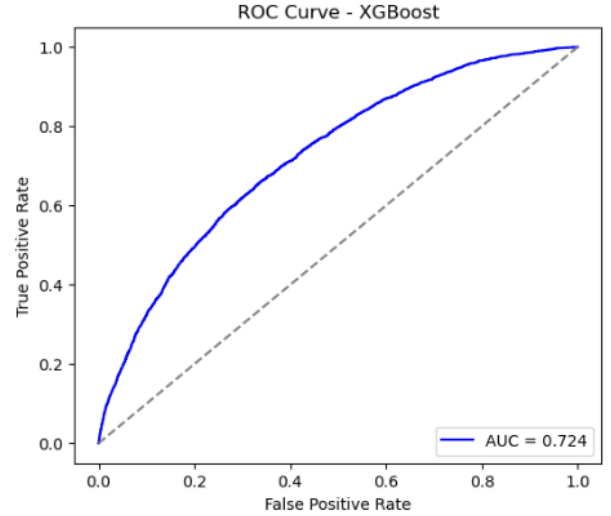
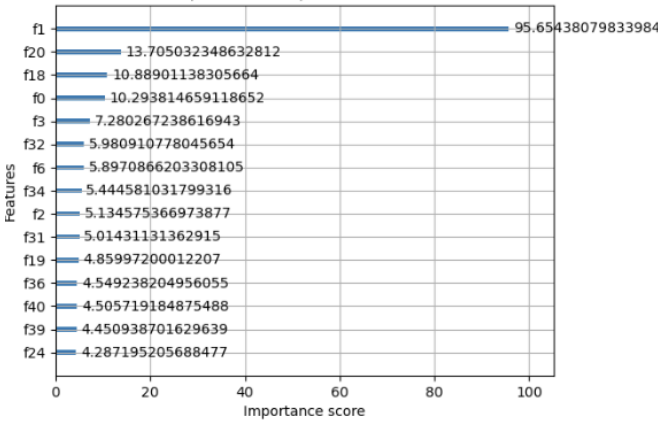


Figure 8 shows the ROC Curve for XGBoost, which evaluates the model's ability to distinguish between classes across different classification thresholds. The curve has an ROC-AUC of 0.724, indicating moderate discriminatory ability between fully paid and charged-off loans. This performance is better than random guessing and provides additional evidence of XGBoost's predictive performance.

Fig. 9. Feature Importance Graph



The Feature Importance Graph shows the relative influence of the variables used in the XGBoost model. The results suggest that a small number of features contribute more strongly to the model than others, while several variables have relatively limited importance. The graph provides a general view of how predictive influence is distributed across the features included in the model.

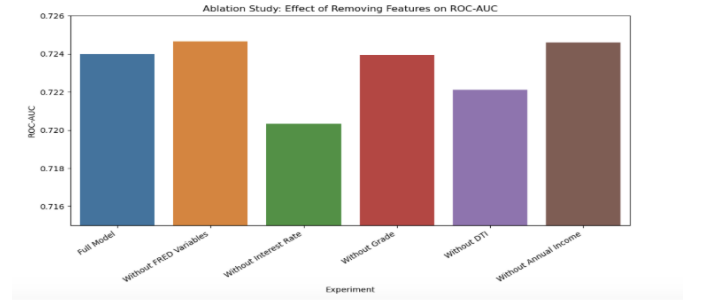
D. Ablation Study

TABLE II
ABLATION STUDY RESULTS

Experiment	Accuracy	Precision	Recall	F1	ROC-AUC
Full Model	0.805	0.576	0.101	0.172	0.724
Without FRED Variables	0.805	0.583	0.104	0.173	0.725
Without Interest Rate	0.804	0.581	0.076	0.135	0.720
Without Grade	0.805	0.579	0.101	0.172	0.724
Without DTI	0.802	0.546	0.076	0.133	0.722
Without Annual Income	0.805	0.574	0.099	0.169	0.725

The goal of the ablation study table is to measure the contribution of selected feature groups. This table yields a few key results. The first main result is that without the FRED variables, ROC-AUC increased slightly from 0.724 to 0.725. This indicates that the macroeconomic variables provided minimal additional predictive value. Additionally, without interest rate, ROC-AUC dropped to 0.720 and F1-score dropped to 0.135, suggesting that interest rate was an important feature in the model. Without loan grade, performance changed very little, indicating a relatively limited contribution. On the other hand, without DTI, ROC-AUC decreased to 0.722, highlighting the importance of borrower indebtedness. Removing annual income increased ROC-AUC by 0.001, indicating that it was a relatively minor factor.

Fig. 10. Ablation Study Graph



The ablation study bar graph shows the performance changes in ROC-AUC when selected features are removed. The graph shows that interest rate has one of the largest impacts, borrower-level variables are generally important, and the macroeconomic variables have only a minimal effect on model performance.

V. DISCUSSION

This study evaluated the performance of the three distinct machine learning models of Logistic Regression, Random Forest, and XGBoost in terms of how well they can predict loan default. In terms of model performance, XGBoost achieved the highest accuracy, precision, and ROC-AUC. It only slightly outperformed the baseline model of Logistic Regression, which shows that the simpler linear models still remain competitive with the more complex ones. In contrast, Random Forest produced the lowest ROC-AUC score of 0.684, making it the weakest of the three models in terms of its ability to distinguish between defaulted and non-defaulted loans. These findings are similar to those in prior literature that found strong predictive performance from XGBoost. However, this study also confirms the advantages of using the Logistic Regression model, as it is the simplest model in terms of interpretability and produces similar results to the XGBoost model. Additionally, through the ablation study, we found that interest rate had the largest impact on the model's performance. On the other hand, removing the FRED variables increased ROC-AUC by only 0.001, suggesting that the macroeconomic variables provided little additional predictive value.

VI. LIMITATIONS AND FUTURE WORK

One limitation of this study is that it only used the LendingClub dataset as well as a smaller macroeconomic dataset containing only eight FRED variables. This means that other macroeconomic variables were omitted that may have shown a more significant change in loan-prediction performance. Another limitation was the class imbalance in the LendingClub dataset, with far more fully paid loans. Because of this, the prediction models would sometimes predict that a loan would be fully paid when it was actually charged off, resulting in low recall and reducing the model's ability to identify defaults. Finally, another limitation is that only three machine learning models were tested. Other literature suggests that LightGBM is another model that could outperform XGBoost and would have been a useful model to include in this study.

VII. CONCLUSION

Machine learning and AI models have become increasingly important in banking and credit lending. While these systems may make lending more efficient and accessible, there also needs to be sufficient governance to ensure that they are used fairly and responsibly. This study evaluated three machine learning models to compare their performance in loan-default prediction. XGBoost achieved the highest accuracy, precision, and ROC-AUC, while Logistic Regression produced similar overall results despite its simpler structure. Additionally, incorporating the FRED macroeconomic variables produced little change in predictive performance, while borrower-level characteristics remained the main drivers of the model.

These findings suggest that XGBoost can provide strong predictive performance for loan-default prediction, although the improvement over Logistic Regression was relatively small. While these small improvements could become meaningful when applied across large numbers of loan decisions, Logistic Regression may still be attractive because of its simplicity and interpretability. Overall, the results suggest that more complex machine learning models can improve credit-risk prediction, but additional model complexity and macroeconomic information do not necessarily lead to large gains in predictive performance.

REFERENCES

- [1] D. West, "Neural network credit scoring models," *Computers & Operations Research*, vol. 27, no. 11–12, pp. 1131–1152, 2000.
- [2] A. Mukesh, "Explainable AI models for credit risk evaluation in banking," *International Journal of Advanced Research in Computer and Communication Engineering*, vol. 13, no. 12, pp. 67–73, 2024.
- [3] J. Mushava and M. Murray, "A novel XGBoost extension for credit scoring class-imbalanced data combining a generalized extreme value link and a modified focal loss function," *Expert Systems with Applications*, vol. 202, p. 117233, 2022.
- [4] M. Tofeeq, "Evaluating the impact of machine learning and alternative data on credit precision and fair lending in emerging markets," *iRASD Journal of Finance*, vol. 3, no. 1, pp. 1–7, 2024.
- [5] J. Li, L. Huang, Z. Jin, and H. Chen, "Study on a prediction of P2P network loan default based on the machine learning LightGBM and XGBoost algorithms according to different high dimensional data cleaning," *Technology and Investment*, vol. 9, no. 4, pp. 173–185, 2018.
- [6] S. Srivastava, "Role of artificial intelligence and alternative data in modern credit scoring models: Transforming lending in India," SSRN, 2026.
- [7] N. Aggarwal, *When All Data Is Credit Data: Consumer Credit Markets, Technological Development, and Distributive Justice*. Oxford, U.K.: University of Oxford, 2023.
- [8] C. Lam, "Debiasing alternative data for credit underwriting using causal inference," arXiv:2410.22382, 2024.
- [9] S. D. Kim, "The double-edged sword of AI and big data: A sociotechnical examination of fairness in alternative lending," arXiv:2307.13408, 2023.
- [10] I. H. Shah, "Investigating the risks of algorithmic bias and explainability failures in credit risk models," SSRN, 2025.